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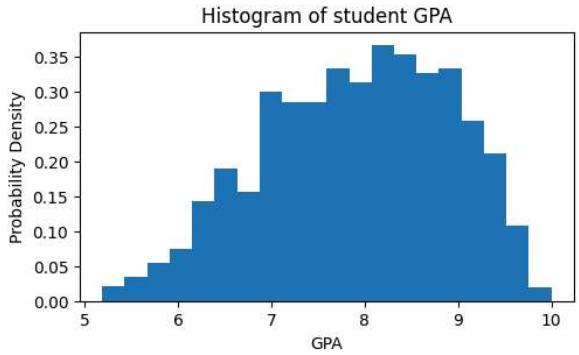
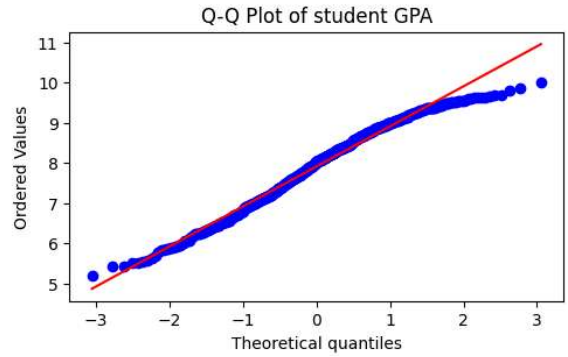
In [9]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import scipy as stats
# load the dataset
data = pd.read_csv('student_gpa.csv')

# display first few rows
print(data.head())
gpa = data['GPA']
# print(gpa)
```

```
      PID             Student Name   GPA
0  232111      ZION SANJAY NARANJE  9.09
1  241001      SHREYA SHAILESH AIL  7.95
2  241002  REHAAN ROSHAN ALEXANDER  8.64
3  241004   SMITH VISHWAS ANDRADES  6.91
4  241005    AADITI SANJAY BARGODE  8.18
```

In [11]:
```
import scipy.stats  # Added import for scipy.stats
plt.figure(figsize=[12,3])
plt.subplot(121)
plt.hist(gpa,bins=20, density=True)
plt.xlabel("GPA")
plt.ylabel("Probability Density")
plt.title("Histogram of student GPA")

plt.subplot(122) # Changed from 121 to 122 for the second subplot
scipy.stats.probplot(gpa, dist="norm", plot=plt) # Changed to scip
plt.title("Q-Q Plot of student GPA")
plt.show()
```





In [ ]:
```
# Comparison of theoretical and actual probailities
# Calulatin Statistial Parameters
total_students = len(gpa)
mean = np.mean(gpa)
std_dev = np.std(gpa, ddof=1)
print("Total students: ", total_students)
print("Mean GPA: ", mean, 4)
print("Standard Deviation: ", round(std_dev,4))
```

In [15]:
```
# Calulatinng Probabilities
pro_theoretial = scipy.stats.norm.cdf(9, mean, std_dev) - scipy.st
prob_actual = ((gpa>=7) & (gpa<=9)).sum()/len(gpa)
print("Theoretical Probability of GPA between 7 and 9: ", round(pr
print("Actual Probability of GPA between 7 and 9: ", round(prob_ac
```

```
Theoretical Probability of GPA between 7 and 9:  0.68
Actual Probability of GPA between 7 and 9:  0.6509
```